

CHAOYI ZHOU

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EDUCATION

Clemson University, US <i>Ph.D in Computer Science</i> Advisor: Prof. Siyu Huang	<i>Jan. 2024 - Present</i>
University of Southern California (USC), US <i>M.S. in Computer Science (General)</i> Advisor: Prof. Yajie Zhao GPA: 3.60 / 4.00	<i>Jan. 2021 - Dec. 2022</i>
Nanjing University Of Posts And Telecommunications (NUPT), China <i>B.S. in Computer Science and Technology</i> GPA: 3.69 / 5.00	<i>Sep. 2016 - Jun. 2020</i>

PUBLICATIONS

Chaoyi Zhou*, Xi Liu*, Feng Luo, Siyu Huang. “Latent Radiance Fields with 3D-aware 2D Representations.” *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

Xi Liu*, **Chaoyi Zhou***, Siyu Huang. “3DGS-Enhancer: Enhancing Unbounded 3D Gaussian Splatting with View-consistent 2D Diffusion Priors.” *Conference on Neural Information Processing Systems (NeurIPS)*, 2024. **[Spotlight]**

RESEARCH EXPERIENCE

Latent Radiance Fields with 3D-aware 2D Representations (ICLR 2025) <i>Graduate Research Assistant</i> <i>Supervisor: Dr. Siyu Huang, Assistant Professor, Clemson University</i> In this work, we propose a method to achieve 3D-aware 2D representations and enable 3D reconstruction in the latent space. Our LRF enables 3D reconstruction on the 2D latent space instead of the image space. It can render high-quality and photorealistic novel views, even for the unbounded scenes.	<i>Jan. 2024 - May. 2024</i>
3DGS-Enhancer (NeurIPS 2024 Spotlight, Top 3.5% of 15,671 submissions) <i>Graduate Research Assistant</i> <i>Supervisor: Dr. Siyu Huang, Assistant Professor, Clemson University</i> Leverage 2D video diffusion priors to tackle the 3D view consistency problem, formulated as temporal consistency in video generation. 3DGS-Enhancer restores view-consistent latent features and integrates them with input views via a spatial-temporal decoder, significantly improving the 3DGS model’s rendering performance.	<i>Jan. 2024 - May. 2024</i>
Outdoor Scene Reconstruction and Understanding <i>Graduate Research Assistant</i> <i>Supervisor: Dr. Rongjun Qin, Associate Professor, Ohio State University</i> Developed visualization software, a Structure-from-Motion system, and a sky camera SDK for reconstructing outdoor scenes and analyzing radiance, enhancing RGB value.	<i>Jun. 2023 - Dec. 2023</i>
Aerial to Ground Novel View Generation <i>Student Researcher, transitioned to Full-Time Computer Vision Engineer</i> <i>Supervisor: Dr. Yajie Zhao, Research Assistant Professor, USC</i> Developed synthetic and real outdoor scene datasets using Unreal Engine 5, and implemented a 3D-flow generative model enhanced with NeRF, applying diffusion models and GANs for high-resolution image inpainting and style transfer.	<i>May 2022 - Jun. 2023</i>

Mechanical System Defect Detection and Tracking

Feb. 2022 - May 2022

Student Researcher

Supervisor: Dr. Preetham Aghalaya Manjunatha, SHM Lab, USC

Implemented multi-process programs and deep learning models for point cloud registration and feature detection, improving precision and localization through transfer learning.

Monocular Depth Estimation and 3D Motion Detection

May. 2021 - Nov. 2021

Research Assistant

Supervisor: Dr. Zhaoxin Li, Institute of Computing Technology, CAS

Developed a multi-scale residual pyramid model for depth and surface normal prediction, and used Structure-from-Motion (SfM) to estimate camera poses and localize 3D targets.

TECHNICAL SKILLS

Programming:	Python, Java, C#, C++
Software & Tools:	Computer Vision: OpenCV, Pointcloud Library (PCL), Open3D
	Machine Learning: PyTorch, TensorFlow, NumPy, Pandas
	Simulation: Unreal Engine, Unity, AirSim

AWARDS

NYIT Presidential and Dean's Honors List, granted by New York Institute of Technology